

AI-Powered Business Intelligence & Data Analytics Project

1. Abstract

This project demonstrates a complete data analytics pipeline from data ingestion to actionable insights using Python and Streamlit. It includes data cleaning, EDA, KPI calculation, trend analysis, driver analysis, forecasting, and AI-generated insights.

2. Introduction

Modern businesses rely on data to make decisions. This project uses the Superstore Sales Dataset to show how raw data is transformed into a Streamlit dashboard that decision-makers can use.

3. Problem Statement & Objectives

Problem: Businesses struggle to turn raw sales data into actionable strategies.

Objectives: Clean data, analyze EDA, compute KPIs, find trends, predict future sales, and generate automated insights.

4. Dataset Description

Dataset: Superstore Sales Data (2011-2015).

Rows: 51,290. Columns: 24.

Key features include Order Date, Sales, Profit, Discount, Category, and Region.

5. System Architecture & Methodology

1. Data Cleaning (Pandas)
2. Analysis Engine (Pandas, NumPy)
3. Forecasting (Scikit-Learn)
4. AI Insights Engine (Rule-based Python)
5. Interactive Dashboard (Streamlit & Plotly)

6. Data Cleaning & EDA

Removed missing values, standardized column names, parsed dates accurately. EDA revealed sales distributions and correlation between discounts and negative profits.

7. KPI, Trend, and Driver Analysis

KPIs (Revenue, Profit, Margin) were dynamically calculated. Driver analysis identified that certain sub-categories and regions drive the most profit, while others operate at a loss due to high discounts.

8. Risk/Opportunity & AI Insights

The Insight Engine flagged specific unprofitable segments as risks and recommended pricing reviews, while

AI-Powered Business Intelligence & Data Analytics Project

highlighting high-margin areas as opportunities for marketing expansion.

9. Forecasting

A Linear Regression model was trained on monthly aggregated sales data. It produced a 6-month forecast with evaluated RMSE and R2 metrics to guide future expectations.

10. Conclusion & Future Scope

The system successfully provides a BI pipeline. Future work includes integrating large language models (LLMs) via APIs for dynamic textual insights and using advanced time-series models like Prophet.